

Management Information Systems

Управување со информациски системи

Chapter 7

Telecommunications, the Internet, and Wireless Technology

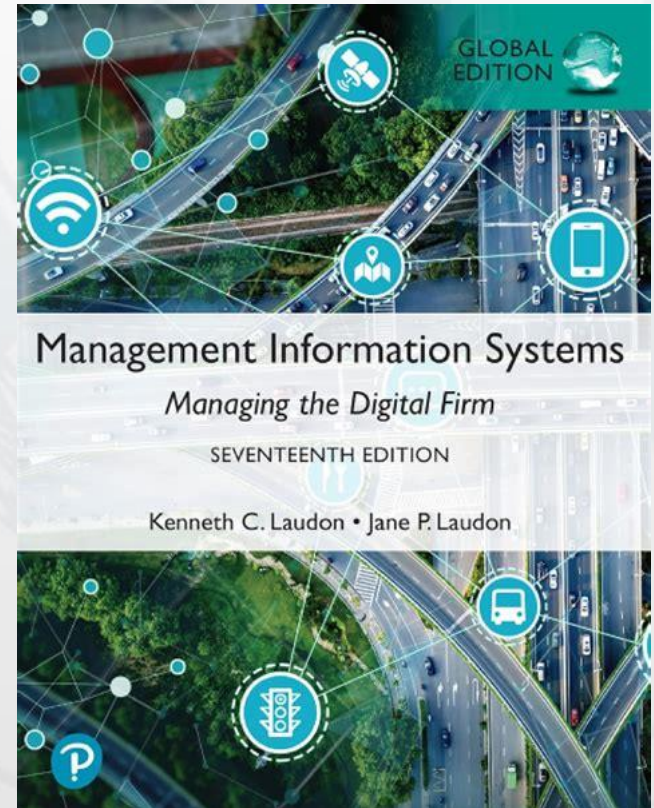
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BOOK

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2022.



Chapter 7

Telecommunications, the Internet, and Wireless Technology



Learning Objectives

- 7.1 What are the principal components of telecommunications networks and key networking technologies?**
- 7.2 What are the different types of networks?**
- 7.3 How do the Internet and Internet technology work, and how do they support communication and e-business?**
- 7.4 What are the principal technologies and standards for wireless networking, communication, and Internet access?**



Networking and Communication Trends

- Convergence
 - Telephone networks and computer networks converging into single digital network using Internet standards
- Broadband
 - Majority of households have high-speed broadband
- Broadband wireless
 - Voice, data communication are increasingly taking place over broadband wireless platforms

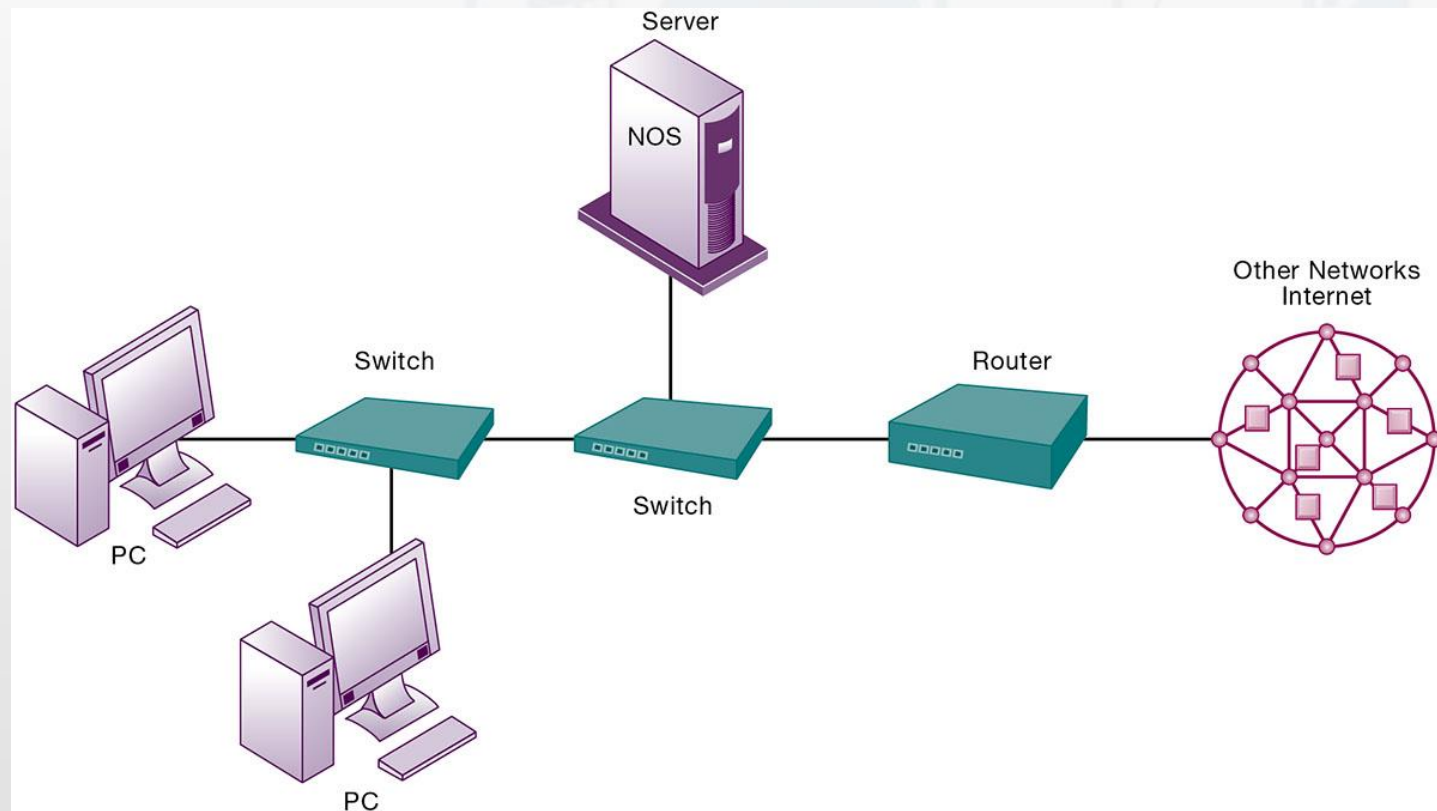


What Is a Computer Network?

- Two or more connected computers
- Major components in simple network
 - Client and server computers
 - Network interfaces (NICs)
 - Connection medium
 - Network operating system (NOS)
 - Hubs, switches, routers
- Software-defined networking (SDN)
 - Functions of switches and routers managed by central program



Figure 7.1 Components of a Simple Computer Network

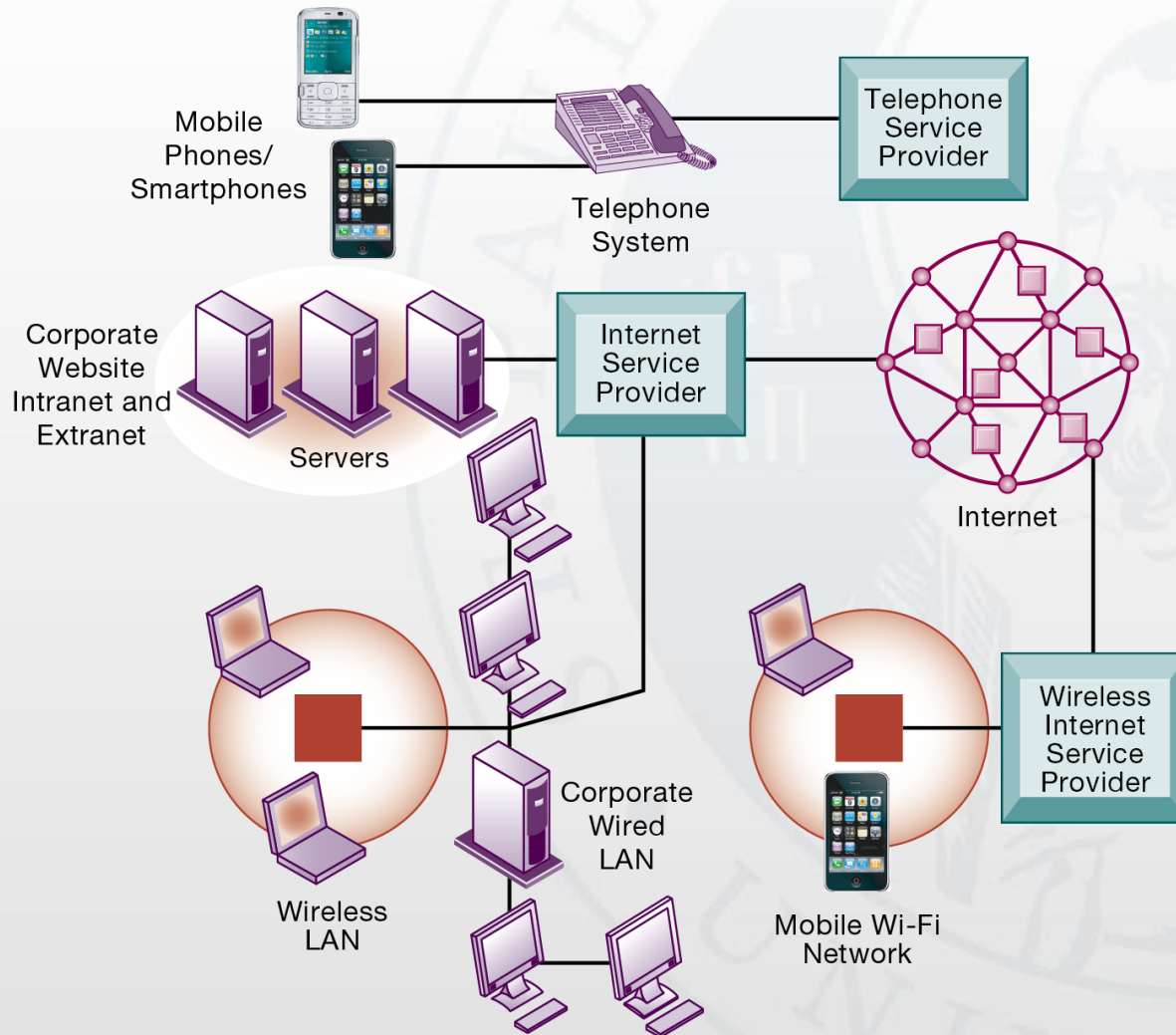


Networks in Large Companies

- Large number of local area networks (LANs) linked to firmwide corporate networks
- Various powerful servers
 - Website, corporate intranet, extranet
 - Backend systems
- Mobile wireless LANs (Wi-Fi networks)
- Videoconferencing system
- Telephone network, wireless cell phones



Figure 7.2 Corporate Network Infrastructure



Key Digital Networking Technologies (1 of 3)

- Client/server computing
 - Distributed computing model
 - Clients linked through network controlled by network server computer
 - Server sets rules of communication for network and provides every client with an address so others can find it on the network
 - Has largely replaced centralized mainframe computing
 - The Internet: largest implementation of client/server computing

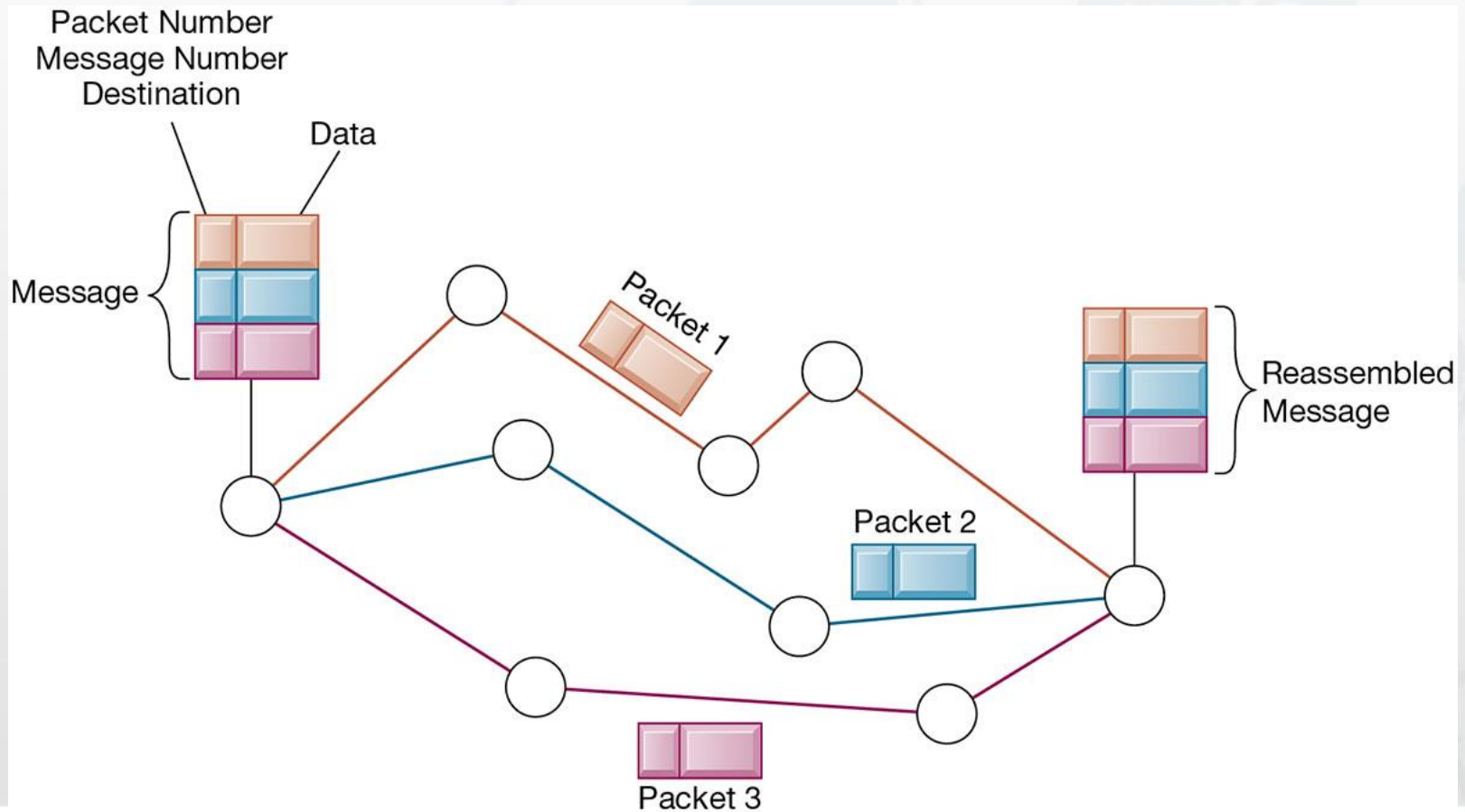


Key Digital Networking Technologies (2 of 3)

- Packet switching
 - Method of slicing digital messages into parcels (packets), sending packets along different communication paths as they become available, and then reassembling packets at destination
 - Previous circuit-switched networks required assembly of complete point-to-point circuit
 - Packet switching more efficient use of network's communications capacity



Figure 7.3 Packet-Switched Networks and Packet Communications

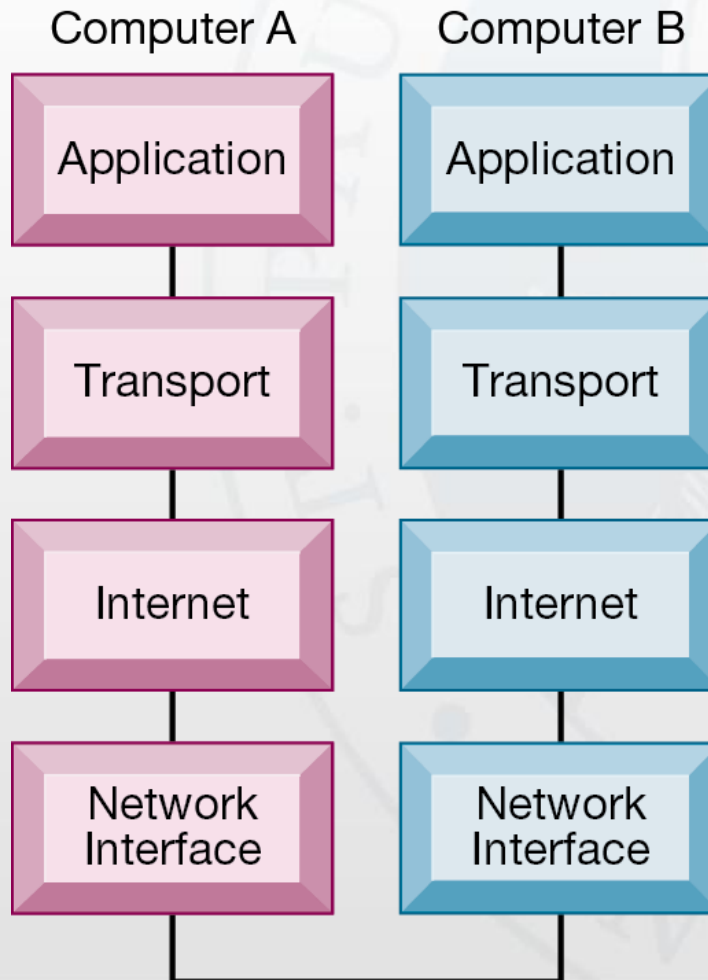


Key Digital Networking Technologies (3 of 3)

- TCP/IP and connectivity
 - Protocols: rules that govern transmission of information between two points
 - Transmission Control Protocol/Internet Protocol (TCP/IP)
 - Common worldwide standard that is basis for the Internet
 - Department of Defense reference model for TCP/IP
 - Four layers
 - Application layer
 - Transport layer
 - Internet layer
 - Network interface layer



Figure 7.4 The Transmission Control Protocol/Internet Protocol (TCP/IP) Reference Model

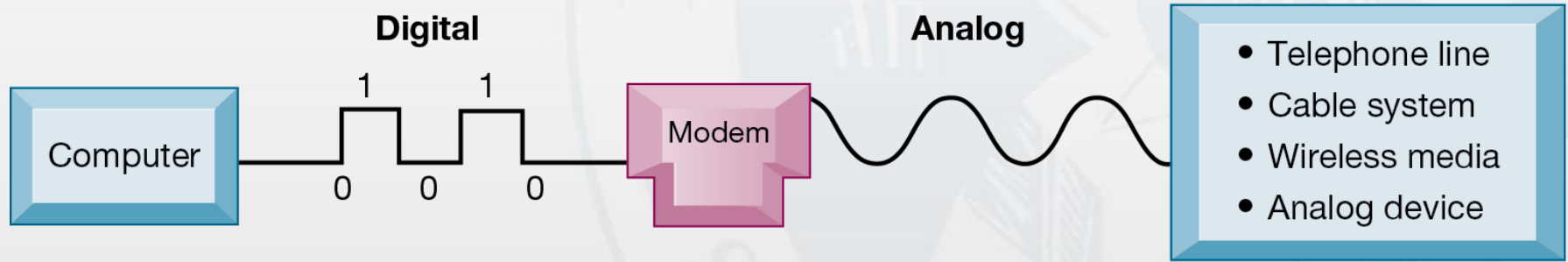


Types of Networks

- Signals: Digital versus analog
 - Modem: translates digital signals into analog form (and vice versa)
- Types of networks
 - Local area networks (LANs)
 - Ethernet
 - Client/server vs. peer-to-peer
 - Wide area networks (WANs)
 - Metropolitan area networks (MANs)
 - Campus area networks (CANs)



Figure 7.5 Functions of the Modem



Transmission Media and Transmission Speed

- Physical transmission media
 - Twisted pair wire (CAT5 and CAT6)
 - Coaxial cable
 - Fiber optics cable
 - Wireless transmission media and devices
 - Satellites
 - Cellular systems
- Bandwidth: Transmission speed
 - Bits per second (bps)
 - Hertz
 - Bandwidth



What Is the Internet?

- The Internet
 - World's most extensive network
 - Internet service providers (ISPs)
 - Provide connections
 - Types of Internet connections
 - Dial-up: 56.6 Kbps
 - Digital subscriber line (DSL/FIOS): 385 Kbps –100+ Mbps
 - Cable Internet connections: 20–100 Mbps
 - Satellite
 - T1/T3 lines: 1.54 Mbps/45 Mbps

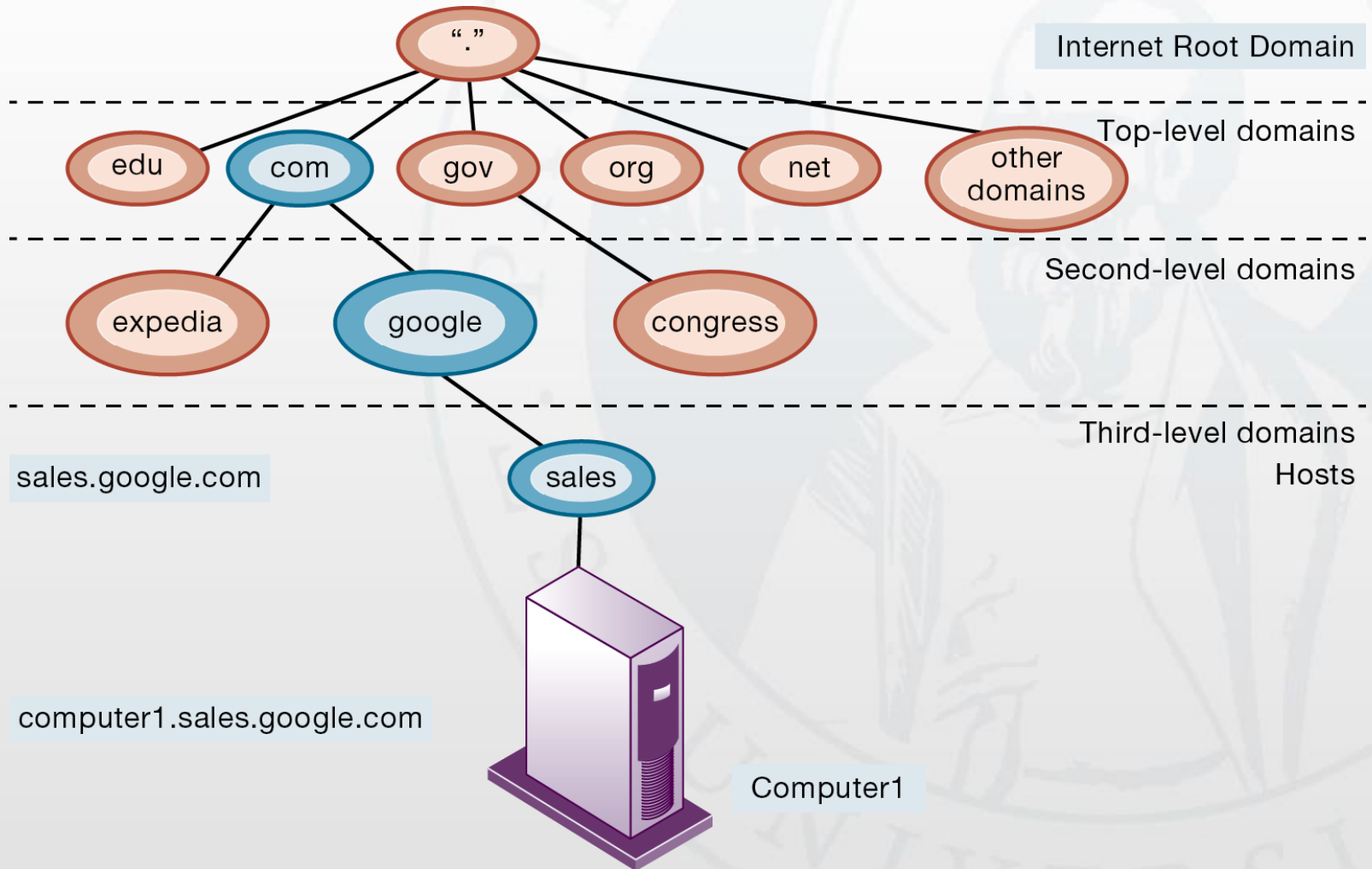


Internet Addressing and Architecture

- Each device on Internet assigned Internet Protocol (IP) address
- 32-bit number, e.g. 207.46.250.119
- The Domain Name System (DNS)
 - Converts IP addresses to domain names
 - Hierarchical structure
 - Top-level domains



Figure 7.6 The Domain Name System

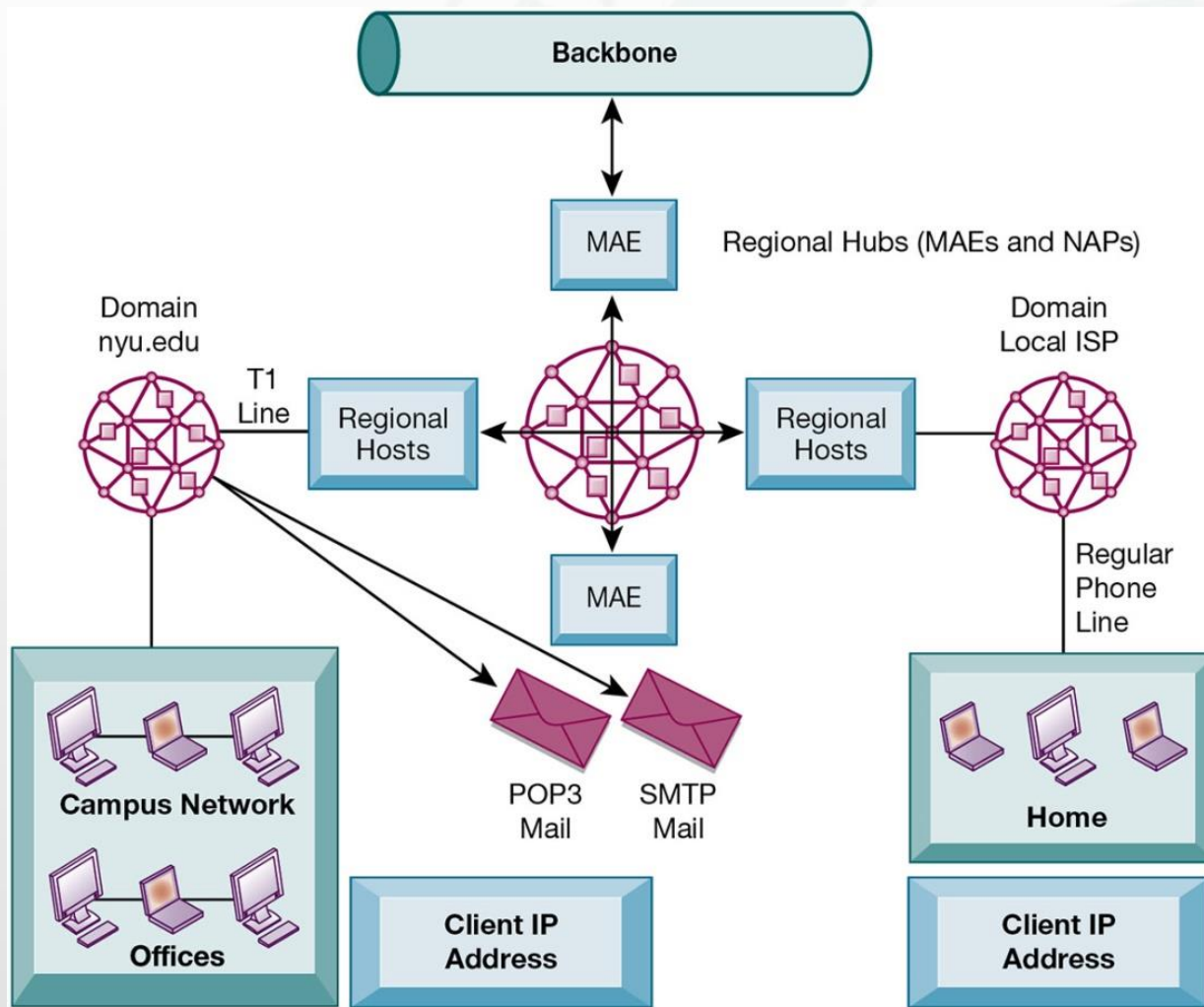


Internet Architecture and Governance

- Network service providers
 - Own trunk lines (high-speed backbone networks)
- Regional telephone and cable T V companies
 - Provide regional and local access
- Professional organizations and government bodies establish Internet standards
 - IAB
 - ICANN
 - W3C



Figure 7.7 Internet Network Architecture



The Future Internet: IPv6 and Internet 2

- IPv6
 - New addressing scheme for IP numbers
 - Will provide more than a quadrillion new addresses
 - Not compatible with earlier IPv4 addressing
- Internet2
 - Advanced networking consortium
 - Universities, businesses, government agencies, other institutions
 - Developed high-capacity 100 Gbps testing network
 - Testing leading-edge new technologies for Internet



Internet Services and Communication Tools (1 of 2)

- Internet services
 - E-mail
 - Chatting and instant messaging
 - Newsgroups
 - Telnet
 - File Transfer Protocol (FTP)
 - World Wide Web

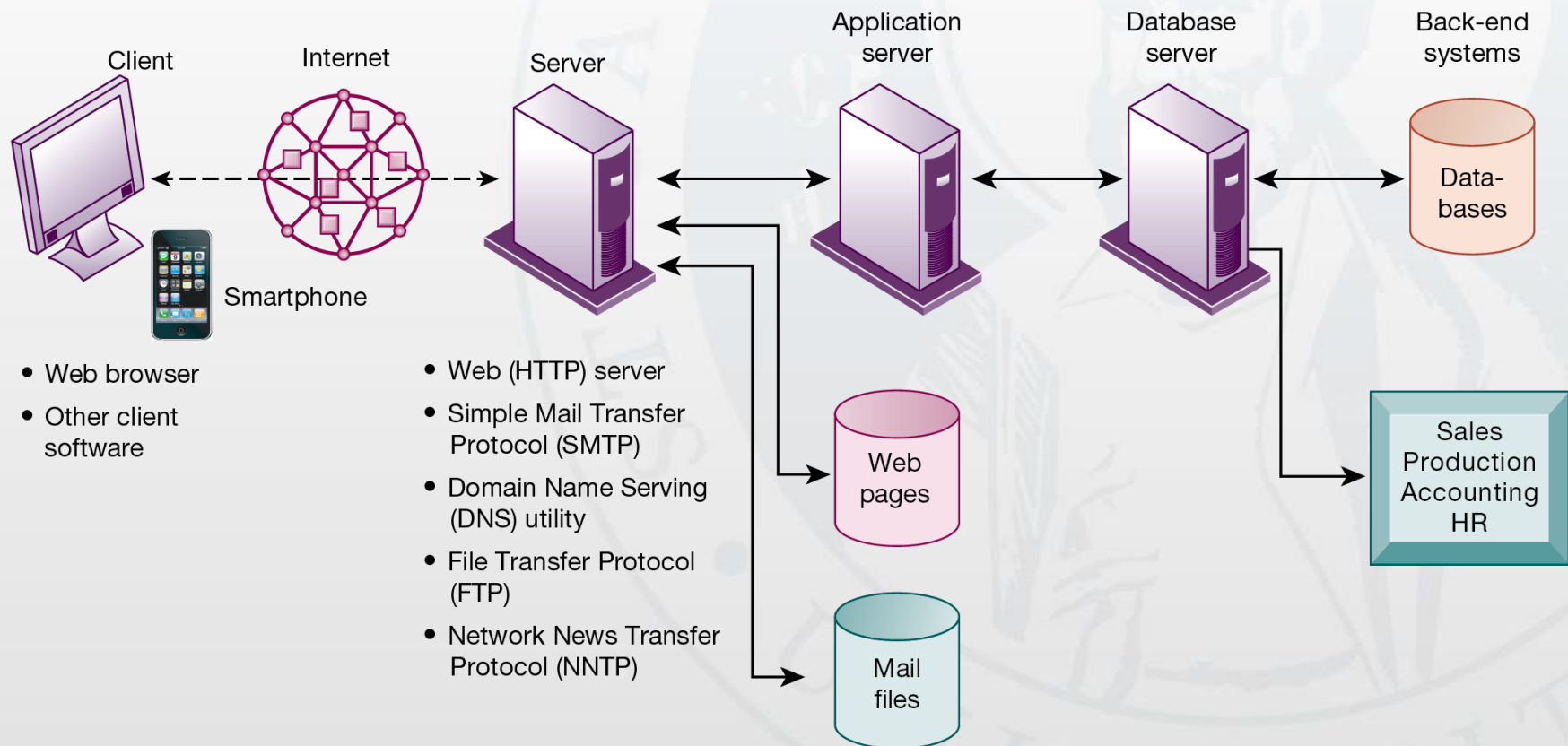


Internet Services and Communication Tools (2 of 2)

- Voice over IP (VoIP)
 - Digital voice communication using IP, packet switching
- Unified communications
 - Communications systems that integrate voice, data, e-mail, conferencing
- Virtual private network (VPN)
 - Secure, encrypted, private network run over Internet



Figure 7.8 Client/Server Computing on the Internet

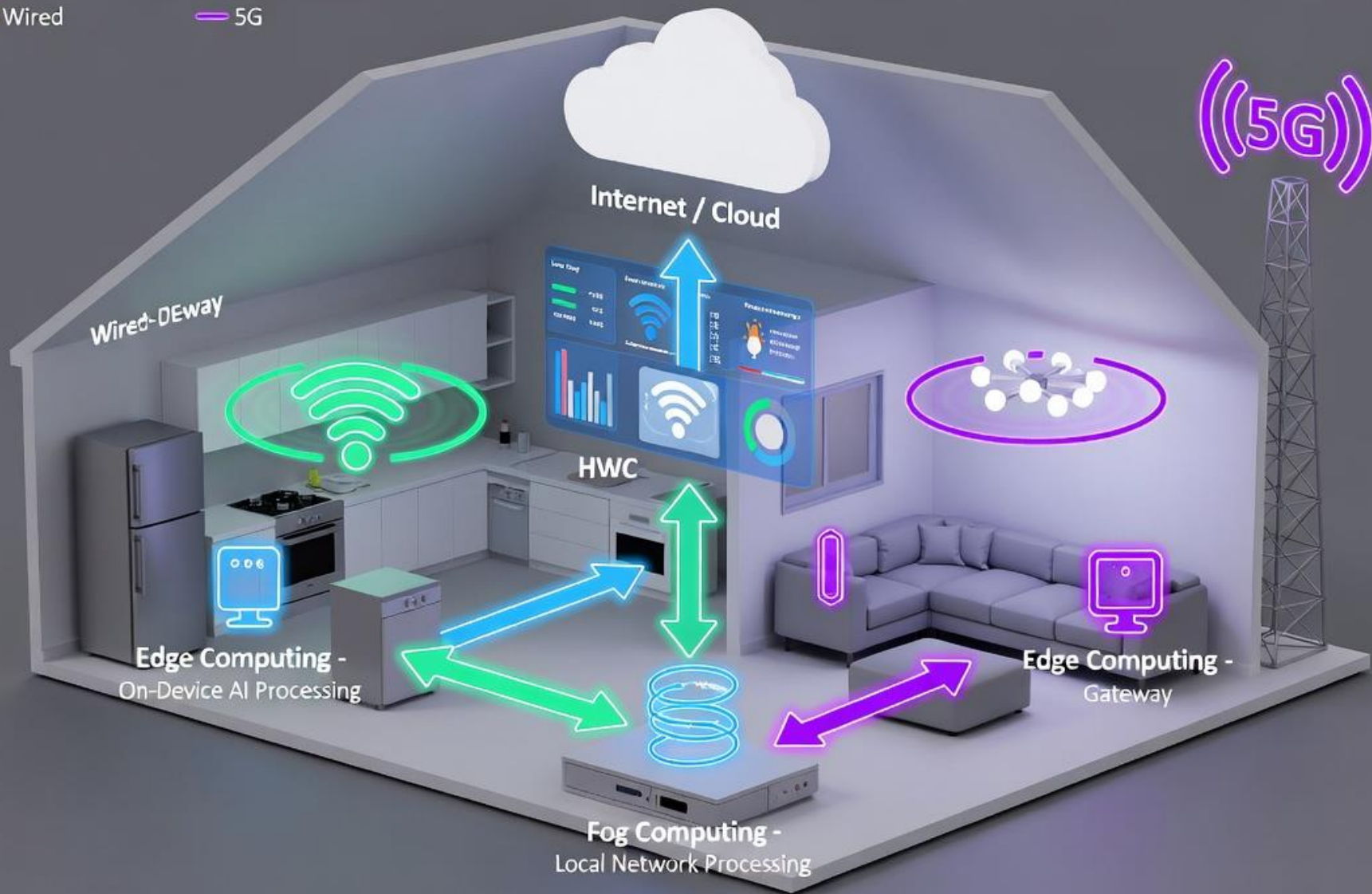


Wired (Ethernet) Wireless Wi-Fi

— Wired

— 5G

Cellular 5G

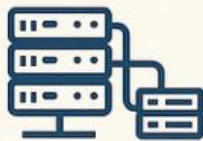


OSS and BSS in telecommunications

OSS vs BSS in Telecom

OSS

OPERATIONS SUPPORT
SYSTEMS



Network
Inventory
Management



Network
Configuration



Service
Provisioning



Fault
Management

BSS

BUSINESS SUPPORT
SYSTEMS



Customer
Relationship
Management (CRM)



Billing
& Charging



Order
Management



Product
Management

- <https://www.youtube.com/watch?v=En-V1DwRCeA>



Figure 7.9 How Voice over IP Works

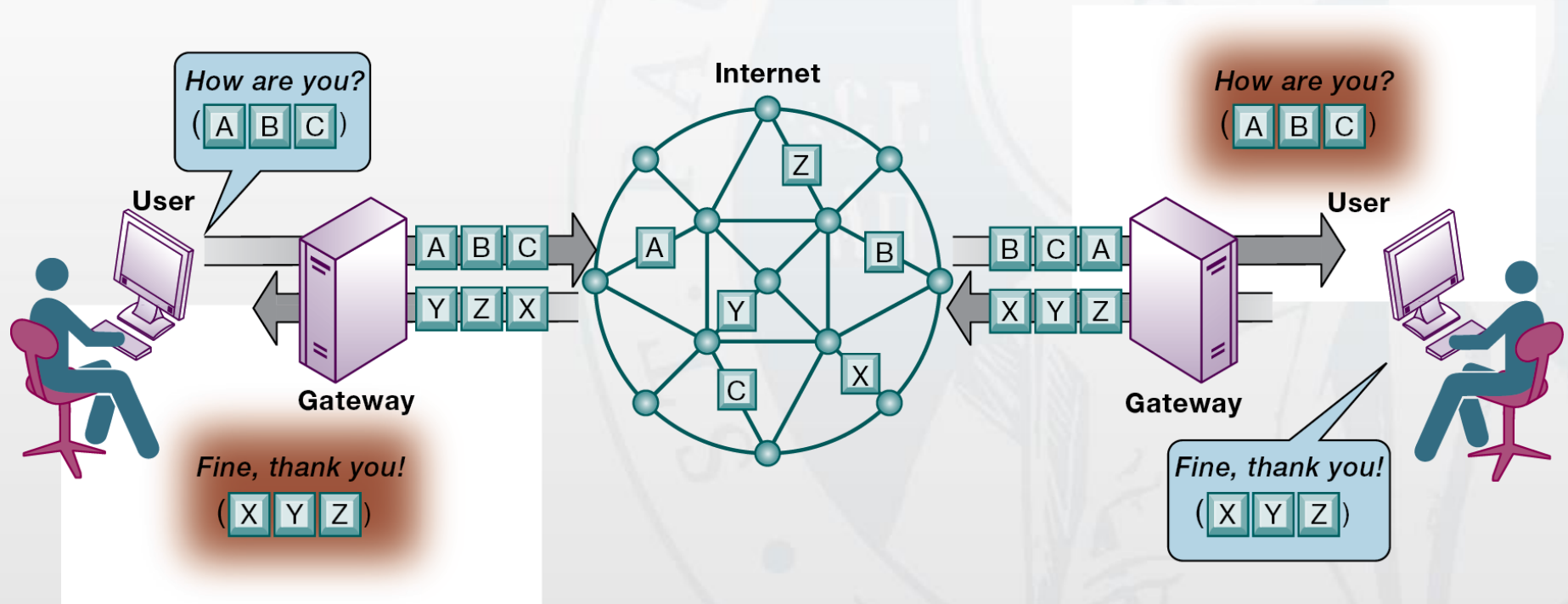
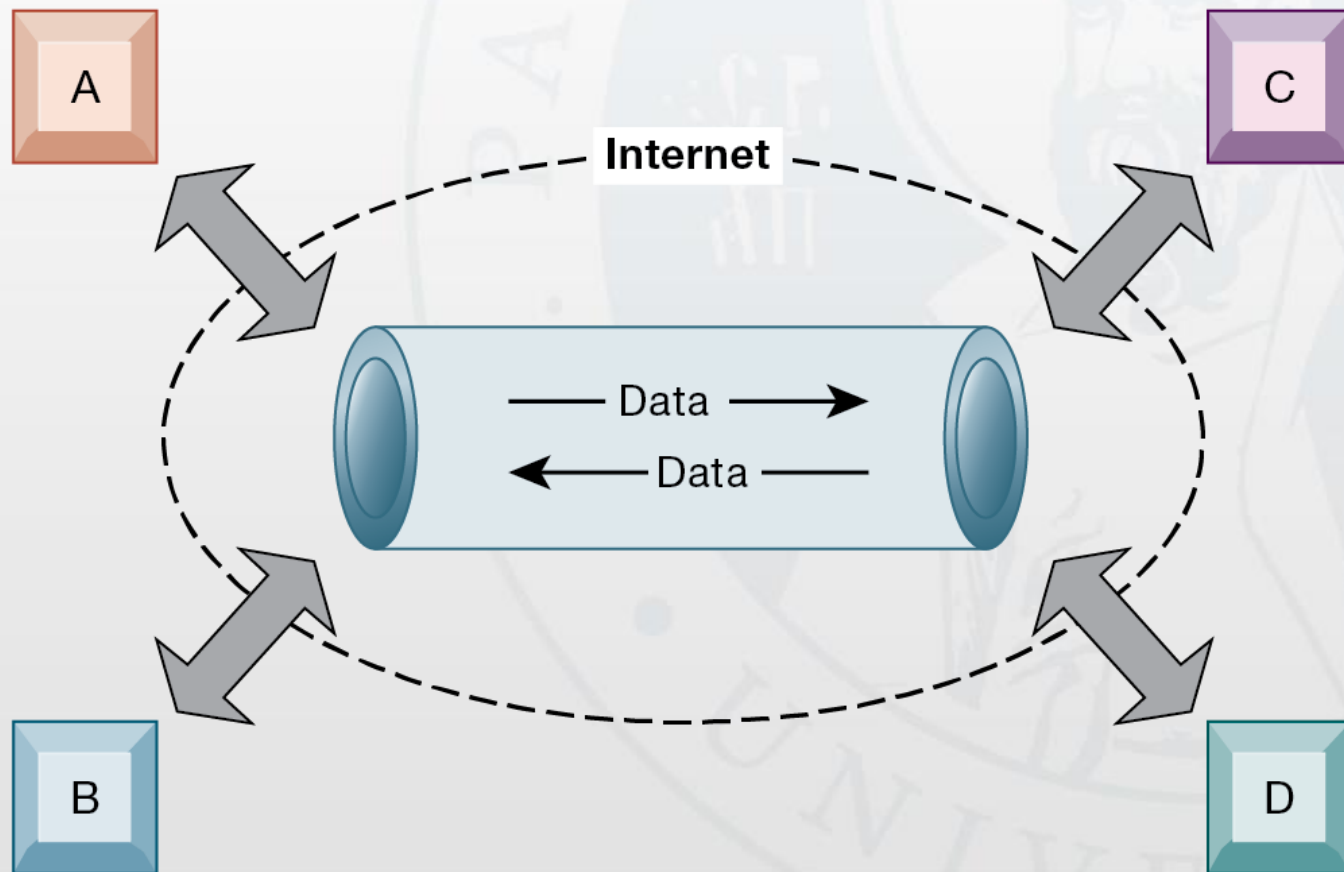


Figure 7.10 A Virtual Private Network Using the Internet



The Web

- Hypertext
 - Hypertext Markup Language (HTML)
 - Hypertext Transfer Protocol (HTTP):
 - Uniform resource locator (URL):
 - <https://uist.edu.mk/info-center/academic-calendar/>
- Web servers
 - Software for locating and managing web pages



Searching for Information on the Web

- Search engines
- Mobile search
- Semantic search and predictive search
- Visual search
- Intelligent agent shopping bots
- Search engine marketing
- Search engine optimization (SEO)



Figure 7.11 Top Desktop/Laptop Web Search Engines Worldwide

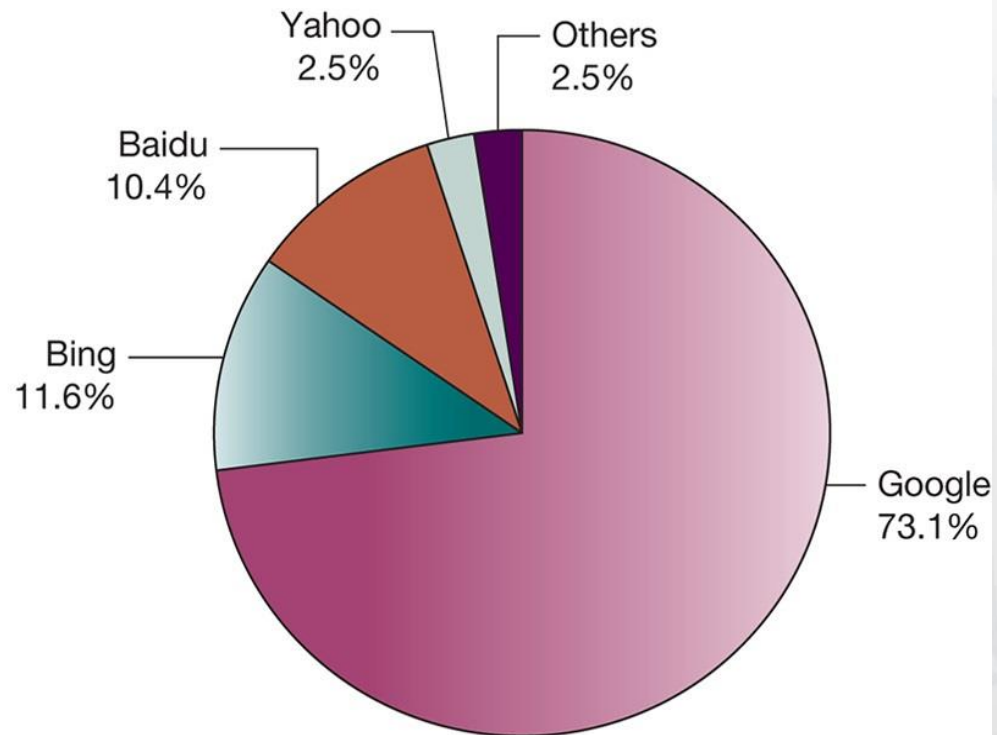
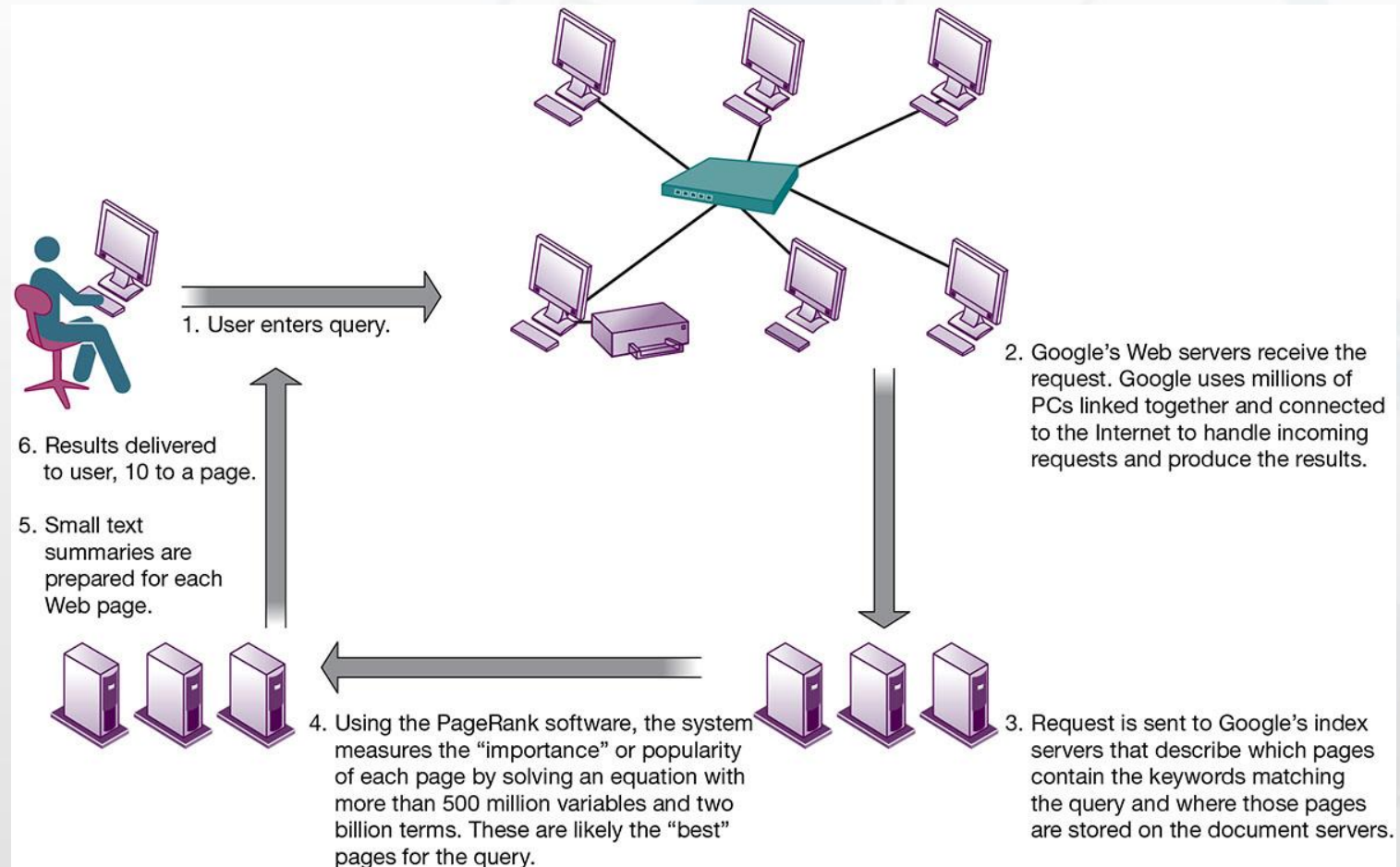


Figure 7.12 How Google Works



Sharing Information on the Web

- Blogs and microblogs
- RSS
- Wikis
- Social networking



The Future Web

- More tools to make sense of trillions of pages on the Internet
- Pervasive web
- Internet of Things (IoT)
- App Internet
- Increased cloud computing and SaaS
- Ubiquitous mobile connectivity
- Greater seamlessness of web as a whole



Cellular Systems (1 of 2)

- Competing standards
 - CDMA : United States only
 - Verizon, Sprint
 - GSM : Rest of world
 - AT&T, T-Mobile
- Third-generation (3G) networks
 - 144 Kbps
 - Suitable for e-mail access, web browsing



Cellular Systems (2 of 2)

- Fourth-generation (4G) networks
 - Up to 100 Mbps
 - Suitable for Internet video
 - LTE and WiMax
- 5G Networks
 - Gigabit capacity
 - Starting to be launched by AT&T, Verizon, and other carriers



Wireless Computer Networks and Internet Access (1 of 2)

- Bluetooth (802.15)
 - Links up to 8 devices in 10-m area using low-power, radio-based communication
 - Useful for personal networking (PANs)
- Wi-Fi (802.11)
 - Set of standards: 802.11
 - Used for wireless LAN and wireless Internet access
 - Use access points: device with radio receiver/transmitter for connecting wireless devices to a wired LAN



Wireless Computer Networks and Internet Access (2 of 2)

- Hotspots: one or more access points in public place to provide maximum wireless coverage for a specific area
- Weak security features
- WiMax (802.16)
 - Wireless access range of 31 miles
 - Require WiMax antennas



Figure 7.13 A Bluetooth Network (PAN)

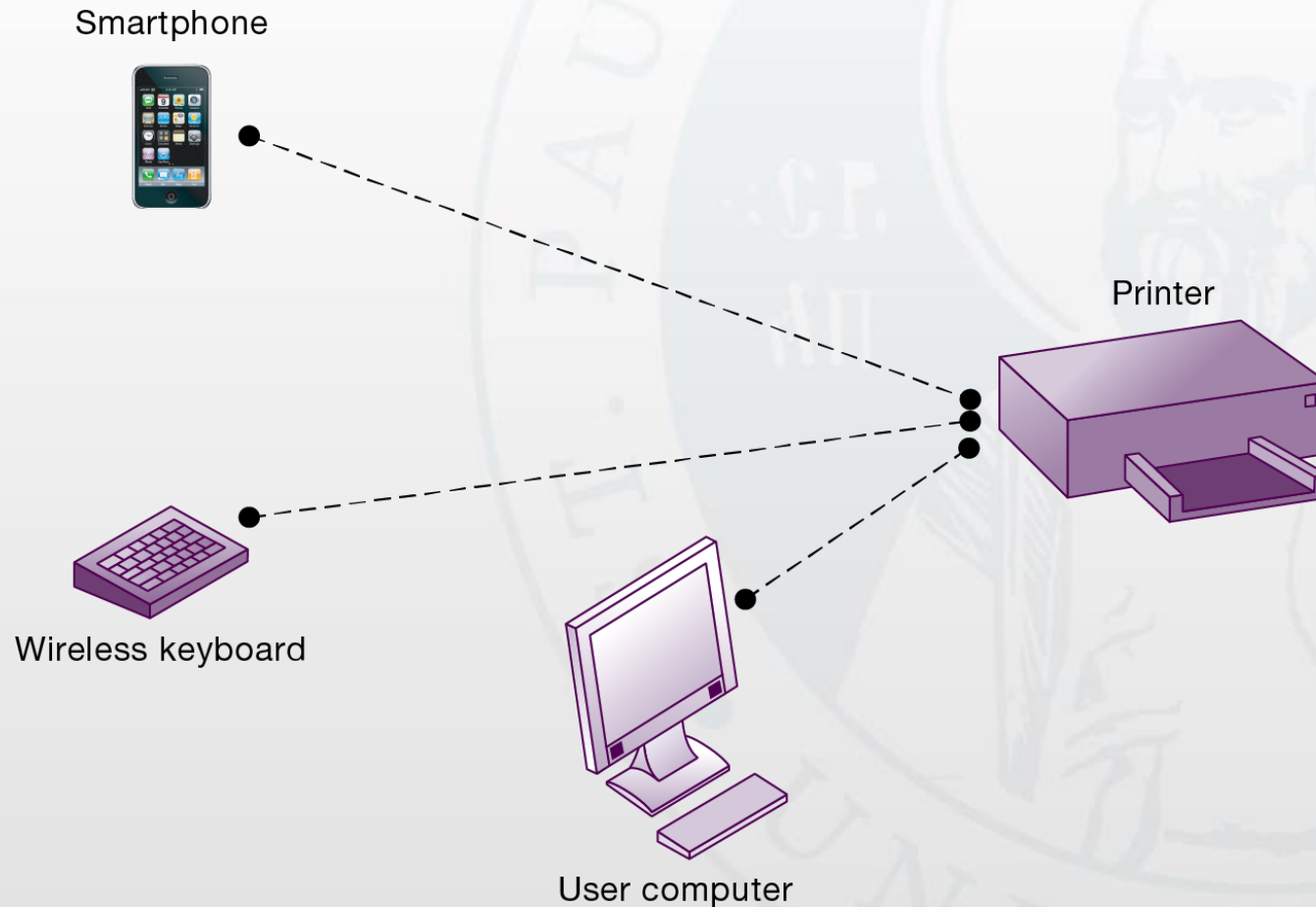
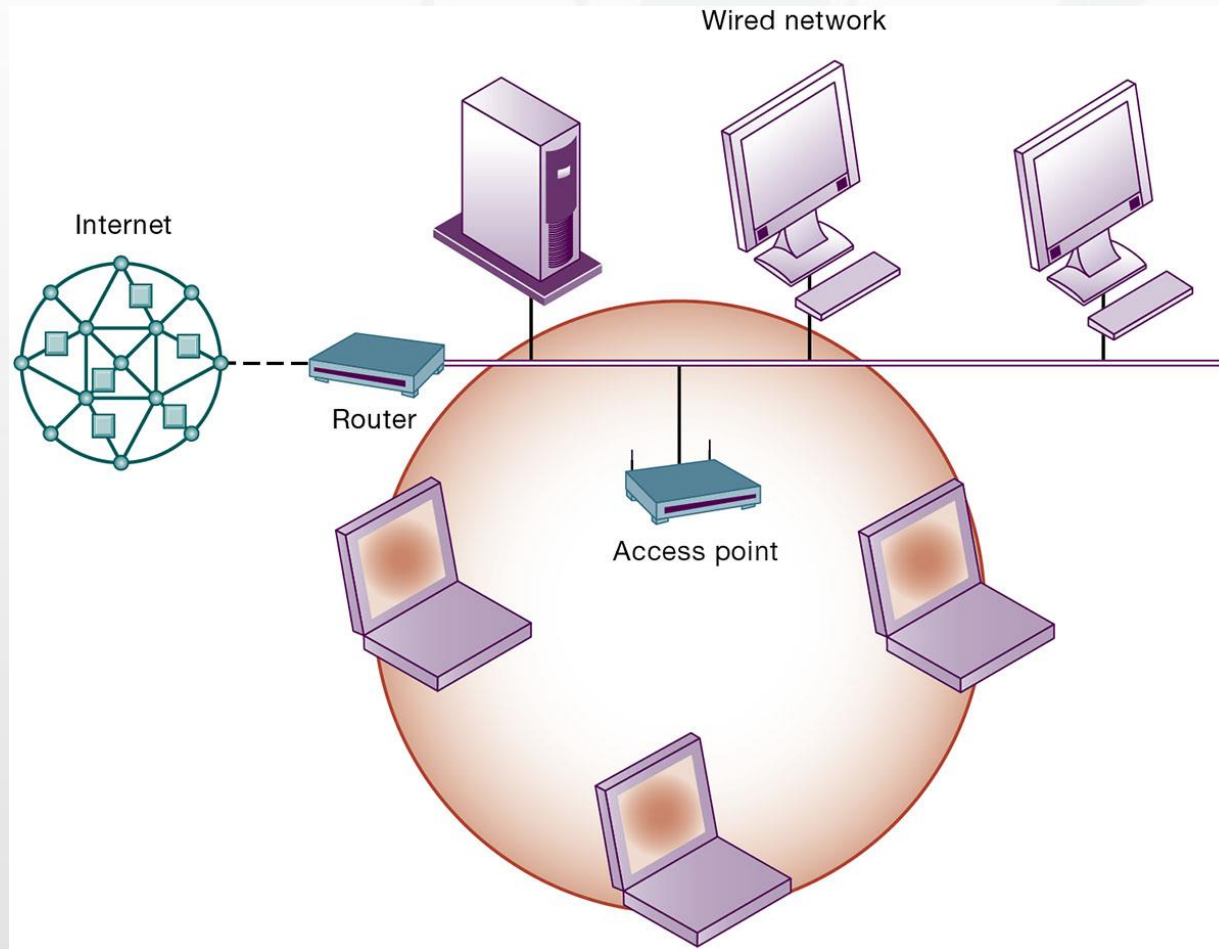


Figure 7.14 An 802.11 Wireless LAN



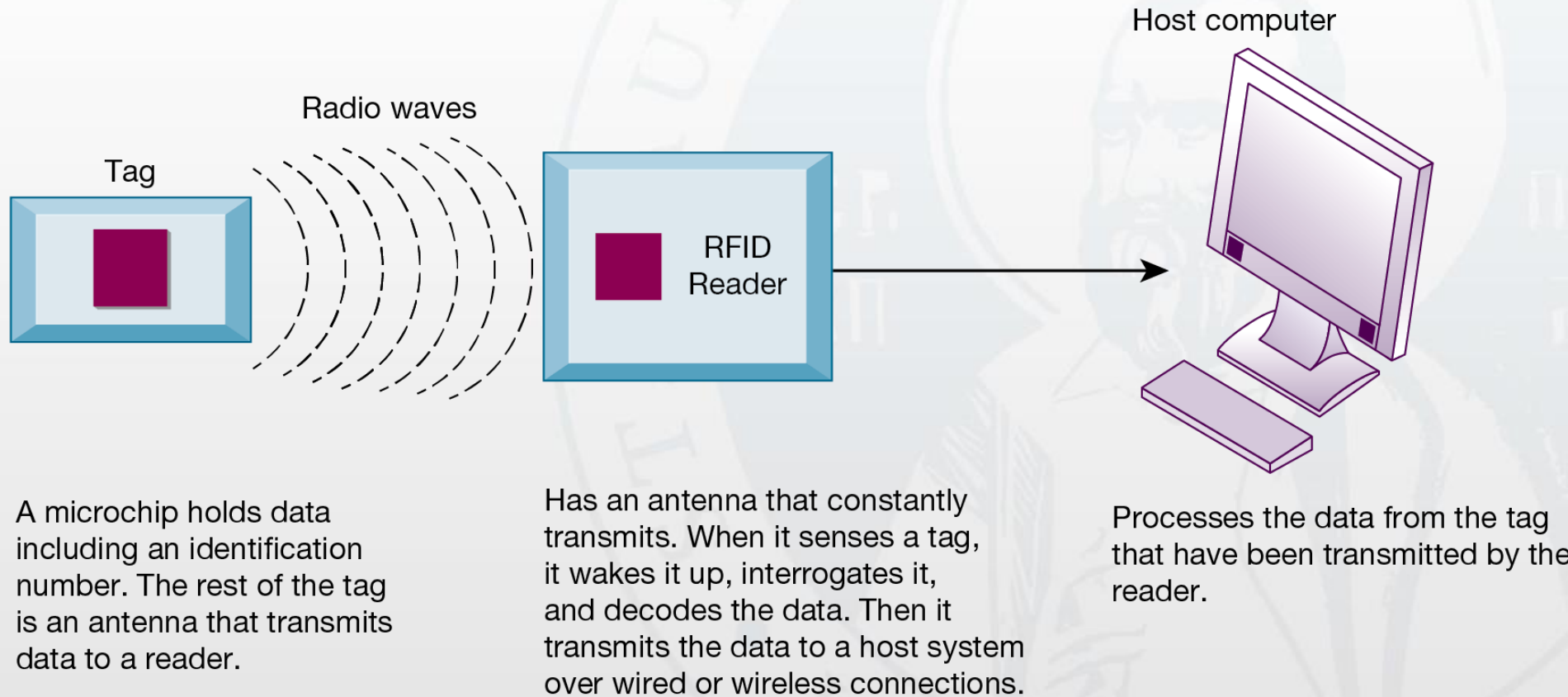
RFID and Wireless Sensor Networks

(1 of 3)

- Radio Frequency Identification (RFID)
 - Use tiny tags with microchips containing data about an item and location; tag antennas to transmit radio signals over short distances to special RFID readers
 - Common uses: Automated toll-collection; tracking goods in a supply chain
 - Reduction in cost of tags making RFID viable for many firms



Figure 7.15 How RFID Works



<https://www.youtube.com/watch?v=qRimFDF7FR0>

RFID and Wireless Sensor Networks (2 of 3)

- Near field communication (NFC)
 - RFID-related technology that uses very short-range wireless connectivity standard
 - Used by tap-and-go services such as Apple Pay, Google Pay

NFC vs. RFID: What's the Difference?

<https://www.youtube.com/watch?v=m0wXeSxQj9I>

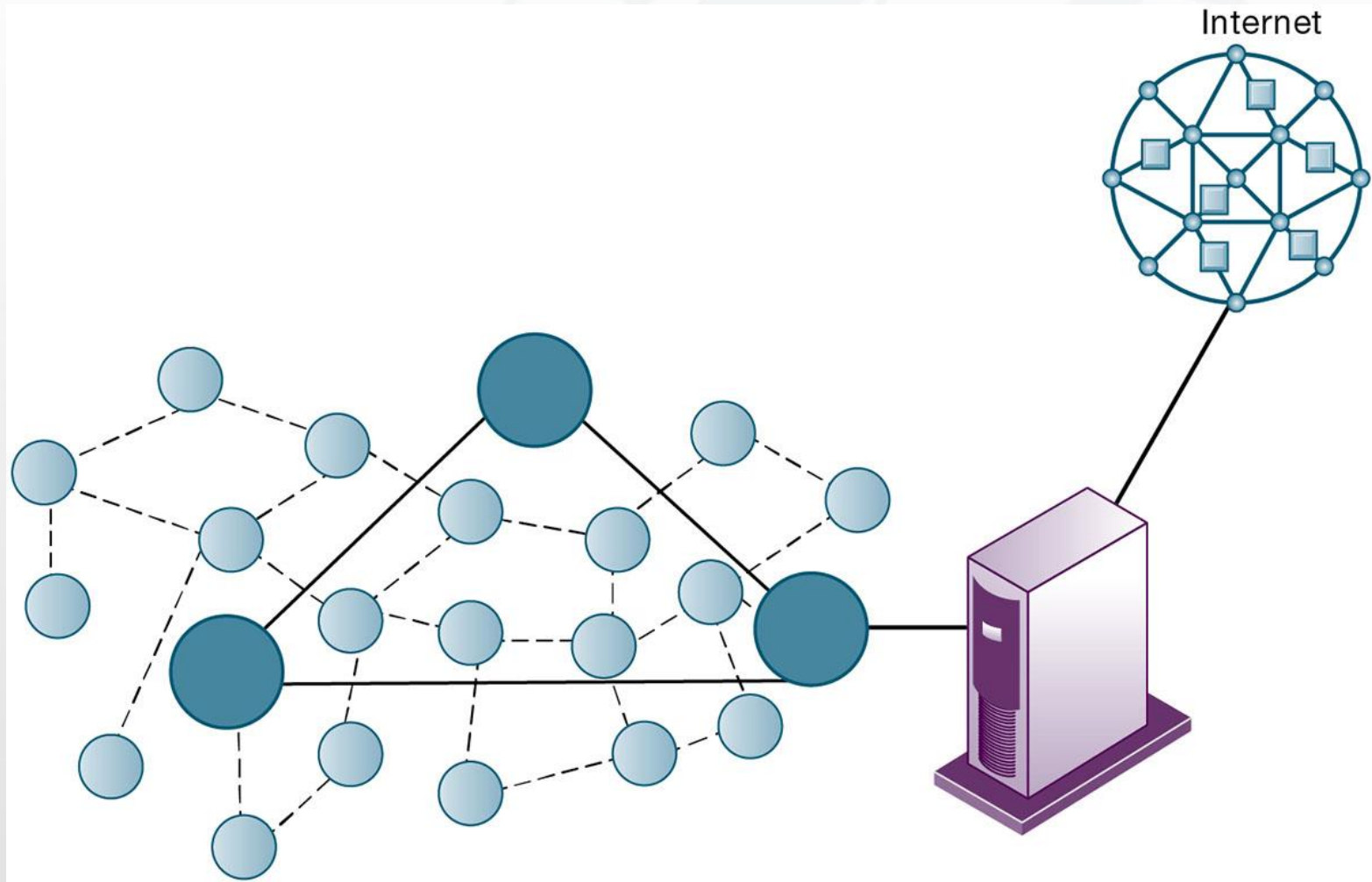


RFID and Wireless Sensor Networks (3 of 3)

- Wireless Sensor Networks (WSNs)
 - Networks of hundreds or thousands of interconnected wireless devices used to monitor building security, detect hazardous substances in air, monitor environmental changes, traffic, or military activity
 - Devices have built-in processing, storage, and radio frequency sensors and antennas
 - Require low-power, long-lasting batteries and ability to endure in the field without maintenance
 - Major sources of big data and fueling Internet of Things



Figure 7.16 A Wireless Sensor Network



Chapter 7 Hands-on MIS Project, pg. 318-319

Using Spreadsheet Software to Evaluate Wireless Services

Software skills: Spreadsheet formulas, formatting

Business skills: Analyzing telecommunications services and costs

You would like to equip your sales force of 35 employees with mobile phones that have capabilities for voice transmission, text messaging, Internet access, and taking and sending photos. Use the web to select two wireless providers that offer nationwide voice and data service as well as good service in your home area. Examine the features of the mobile handsets and wireless plans offered by each of these vendors.. Use your spreadsheet software to determine the mobile service and handset that will offer the best pricing per user over a two-year period. For the purposes of this exercise, you do not need to consider corporate discounts.



Chapter 7 Hands-on MIS Project, pg. 318-319

Using Web Search Engines for Business Research

Software skills: Web search tools

Business skills: Researching new technologies

This project will help develop your Internet skills in using web search engines for business research.

Use Google and Bing to obtain information about ethanol as an alternative fuel for motor vehicles. If you wish, try some other search engines as well.

Compare the volume and quality of information you find with each search tool. Which tool is the easiest to use?

Which produced the best results for your research? Why?



Management Information Systems

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